Math 105L **Lab 4 Part 1**

Human Growth: Computing Rates of Change from Age 3 to 20

1. Background & Height Data Overview

In this laboratory, we investigate longitudinal growth data tracking the median stature of US children from age 3 to 20 (36 to 240 months):

- **Height Graphs $H_M(t)$ and $H_F(t)$:** Both curves increase monotonically across childhood. From the raw height curves alone, male and female trajectories appear very similar until age 13, after which female growth flattens while male growth continues.
- **Calculating Growth Rates:** Because data points are recorded monthly ($\Delta t = 1$ month $= \frac{1}{12}$ year), the annual growth rate in cm/year is calculated via:

$$\text{Growth Rate } \mathbf{H'(t)} \approx \frac{\mathbf{H(t+\Delta t) - H(t)}}{\frac{\mathbf{1}}{\mathbf{12}}} = \mathbf{12} \cdot \big[\mathbf{H(t+1) - H(t)}\big]$$

What the Growth Rate Curves Reveal

- **Early Childhood Deceleration (Ages 3–8):** Growth rates start high ($\approx$ 7.5–8 cm/year) and steadily decrease as infant growth slows down.
- **Pre-Spurt Minimum (Ages 9–11):** Growth reaches its lowest childhood rate ($\approx$ 5 cm/year) just prior to the onset of puberty.
- **Adolescent Growth Spurt Peaks:**
  - **Females:** Peak growth rate occurs at approximately **140 months ($\approx$ 11.7 years)**, reaching $\approx$ 7.5 cm/year.
  - **Males:** Peak growth rate occurs roughly 1.5–2 years later, at approximately **160 months ($\approx$ 13.3 years)**, reaching a higher peak of $\approx$ 8.2 cm/year.
- **Post-Spurt Cessation:** Both curves decline sharply toward 0 cm/year as growth plates fuse and adult stature is attained.

2. Fundamental Calculus Insights

- **Sign of $H'(t)$ vs. Change in $H'(t)$:** While $H'(t) > 0$ for all ages (children never shrink), the derivative itself is not constant. It decreases, increases, and decreases again.
- **Rate Peaks as Inflection Points:** The peak of a growth spurt represents the instant of **fastest growth** (maximum rate). On the stature graph $H(t)$, this corresponds to the point where the curve is **steepest**—an inflection point transitioning from concave up to concave down!